

EXPLAINABLE MACHINE LEARNING FRAMEWORK FOR CHRONIC KIDNEY DISEASE PREDICTION USING SHAP-BASED INTERPRETABILITY

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*Research Interest: Explainable AI,
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ABSTRACT

Artificial Intelligence and Machine Learning are increasingly being adopted in healthcare systems for disease prediction and clinical decision support. Although machine learning models can achieve strong predictive performance, many healthcare AI systems still operate as black-box models that lack interpretability. In healthcare environments, clinicians require not only accurate predictions but also transparent explanations behind those predictions.

This research presents an Explainable Artificial Intelligence (XAI) framework for Chronic Kidney Disease (CKD) prediction using Machine Learning and SHAP (SHapley Additive Explanations). The study utilizes the Chronic Kidney Disease dataset from the UCI Machine Learning Repository containing 400 patient records with multiple clinical attributes related to kidney function assessment.

The proposed framework applies a Decision Tree Classifier integrated with SHAP-based explainability analysis to improve transparency and interpretability in healthcare prediction systems. Data preprocessing techniques including missing value handling, categorical

encoding, and numerical transformation were performed prior to model development.

Experimental results demonstrated strong predictive performance with high accuracy, ROC-AUC score, and cross-validation performance. SHAP explainability analysis identified clinically significant features such as Hemoglobin, Specific Gravity, Blood Urea, Serum Creatinine, and Hypertension as major contributors influencing CKD prediction outcomes.

The findings demonstrate that combining predictive machine learning models with explainability techniques can improve trust, transparency, and interpretability in AI-assisted healthcare systems.

Keywords— Explainable AI, Machine Learning, SHAP, Chronic Kidney Disease, Healthcare Analytics, Clinical Prediction, Decision Tree, Interpretable Machine Learning

1. INTRODUCTION

Artificial Intelligence and Machine Learning are increasingly being adopted in healthcare systems for disease prediction, clinical risk assessment, and medical decision-support systems. Predictive healthcare models can analyze patient health records and identify patterns associated with disease progression. Chronic Kidney Disease (CKD) has become a major global healthcare concern due to its increasing prevalence and long-term health complications.

Early detection of CKD is critical because delayed diagnosis may lead to kidney failure, cardiovascular complications, and increased mortality risk. Machine learning approaches can assist healthcare professionals by enabling early-stage prediction using clinical biomarkers and patient health information.

Despite the growing use of machine learning in healthcare, many predictive models operate as black-box systems where the internal reasoning

behind predictions remains unclear. In healthcare environments, predictive accuracy alone is insufficient because clinicians must understand why a model generates a particular prediction before integrating it into medical practice.

Explainable Artificial Intelligence (XAI) techniques address this challenge by improving transparency and interpretability in AI systems. SHAP (SHapley Additive Explanations) is one of the most widely used explainability frameworks because it provides both local and global explanations regarding feature contribution and model behavior.

This research focuses on developing an explainable machine learning framework for Chronic Kidney Disease prediction using Decision Tree classification integrated with SHAP interpretability analysis.

2. OBJECTIVES

- To analyze the Chronic Kidney Disease healthcare dataset.
- To preprocess and clean clinical healthcare data.
- To implement a Decision Tree classifier for CKD prediction.
- To evaluate model performance using standard classification metrics.
- To apply SHAP explainability techniques for model interpretation.
- To identify clinically important features contributing to CKD prediction.
- To improve transparency and trust in healthcare AI systems.

3. RELATED WORK

Machine learning has been widely applied in healthcare analytics for disease prediction and diagnosis support. Previous studies have explored algorithms such as Logistic

Regression, Decision Trees, Random Forests, and Support Vector Machines for Chronic Kidney Disease prediction.

Recent healthcare AI research has emphasized the importance of model interpretability due to the limitations of black-box systems in clinical environments. Explainable AI approaches including SHAP and LIME have emerged as important techniques for improving transparency in machine learning predictions.

Several studies have shown that explainability frameworks can assist clinicians in understanding feature-level contributions influencing disease outcomes. SHAP-based explanations are particularly useful because they provide consistent and theoretically grounded feature importance analysis.

This study extends existing healthcare AI research by integrating Decision Tree classification with SHAP explainability analysis for interpretable CKD prediction.

4. DATASET DESCRIPTION

The Chronic Kidney Disease dataset used in this study was obtained from the UCI Machine Learning Repository. The dataset contains clinical and laboratory records collected from patients for CKD diagnosis analysis.

Dataset Overview

- Total Instances: 400 patients
- Total Features: 25 clinical attributes
- Task Type: Binary Classification
- Target Variable: CKD / Not CKD

Important Clinical Features

- Hemoglobin (hemo)
- Specific Gravity (sg)
- Blood Urea (bu)
- Serum Creatinine (sc)
- Packed Cell Volume (pcv)
- Hypertension (htn)
- Red Blood Cell Count (rc)

- Potassium (pot)

Target Variable

- ckd → Chronic Kidney Disease
- notckd → No Chronic Kidney Disease

5. METHODOLOGY

5.1 Data Collection

The CKD dataset was imported into the Python analysis environment using Pandas.

5.2 Data Preprocessing

The preprocessing stage included:

- Removal of unnecessary identifier columns
- Conversion of categorical variables into numerical form
- Missing value handling using SimpleImputer
- Feature transformation
- Train-test splitting for model training and evaluation

5.3 Machine Learning Model

A Decision Tree classifier was implemented using Scikit-learn.

The model was trained using:

- Training Dataset: 80%
- Testing Dataset: 20%

5.5 Explainability Analysis

SHAP explainability analysis was performed to understand feature-level contributions influencing CKD prediction outcomes.

SHAP analysis enabled:

- Global model interpretation
- Local prediction explanation
- Feature importance analysis
- Transparent healthcare AI interpretation

6. MACHINE LEARNING IMPLEMENTATION

The Decision Tree classifier was trained using the pre-processed CKD dataset. The model successfully learned relationships between clinical variables and disease outcomes.

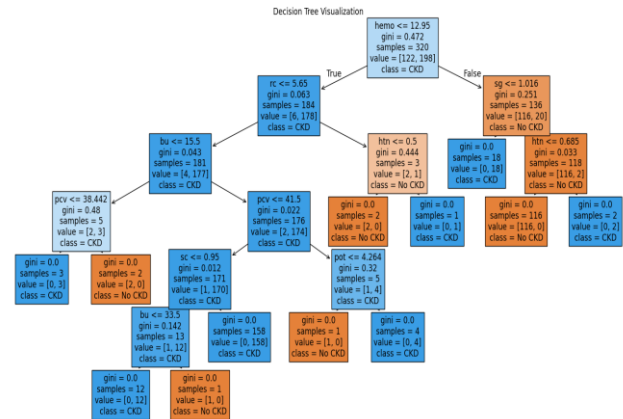


Fig 1: Decision Tree structure used for CKD prediction

8. RESULTS AND PERFORMANCE ANALYSIS

The model was evaluated using multiple classification metrics.

Performance Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Cross Validation

Metric	Score
Accuracy	1.00
Precision	1.00
Recall	1.00
F1-Score	1.00

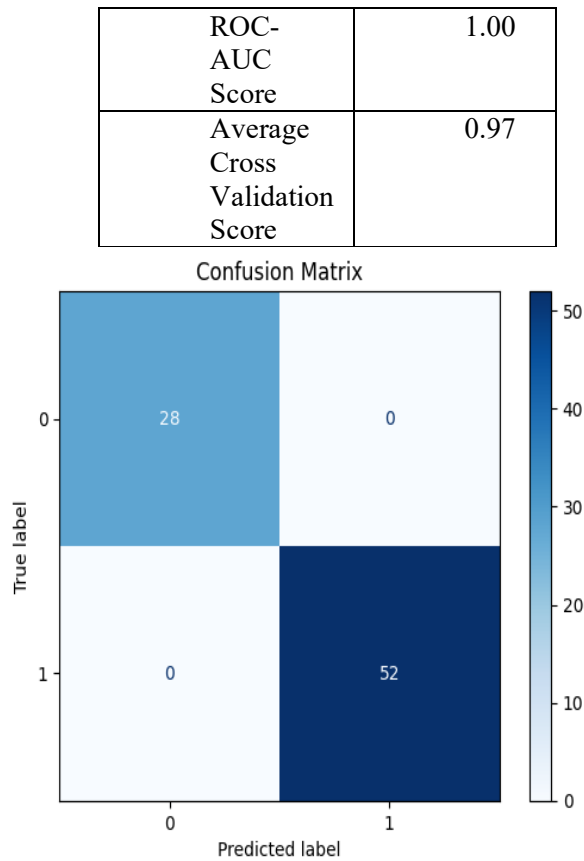


Fig 2: Confusion Matrix for Decision Tree classification model

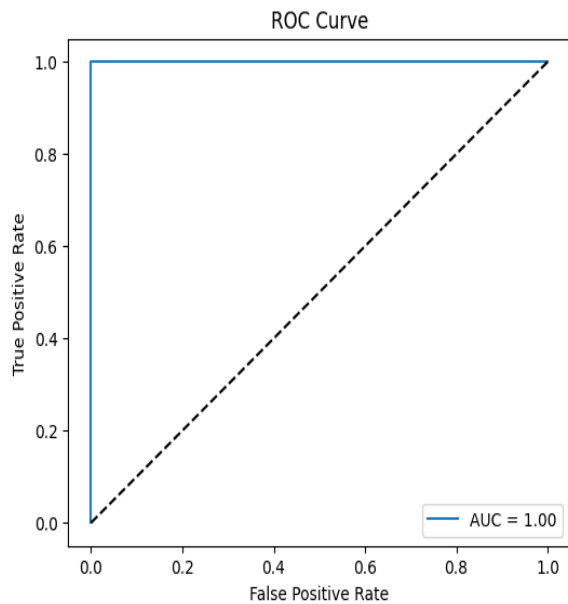


Fig 3: ROC Curve showing classification performance of the Decision Tree model

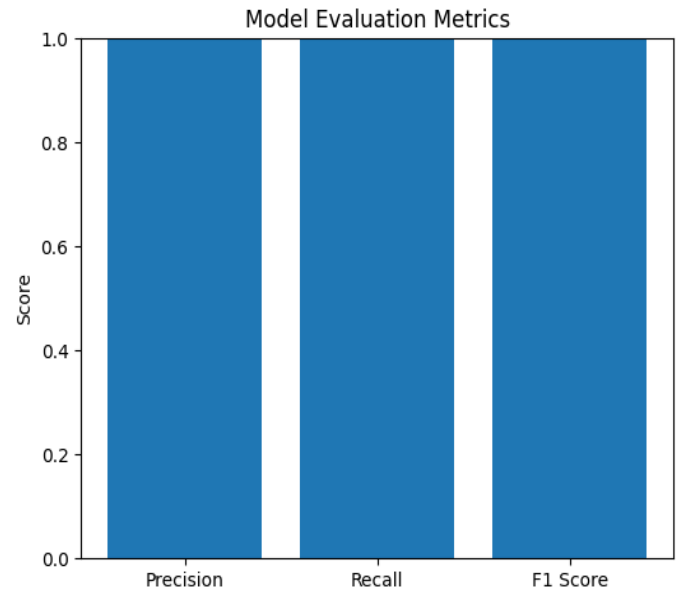


Fig 4: Evaluation metrics comparison for the proposed model

Observation:

The results demonstrate strong predictive performance for CKD classification.

8. SHAP EXPLAINABILITY ANALYSIS

SHAP analysis was performed to identify clinically significant features influencing CKD prediction outcomes.

Important Features Identified:

- Hemoglobin (hemo)
- Specific Gravity (sg)
- Hypertension (htn)
- Blood Urea (bu)
- Packed Cell Volume (pcv)

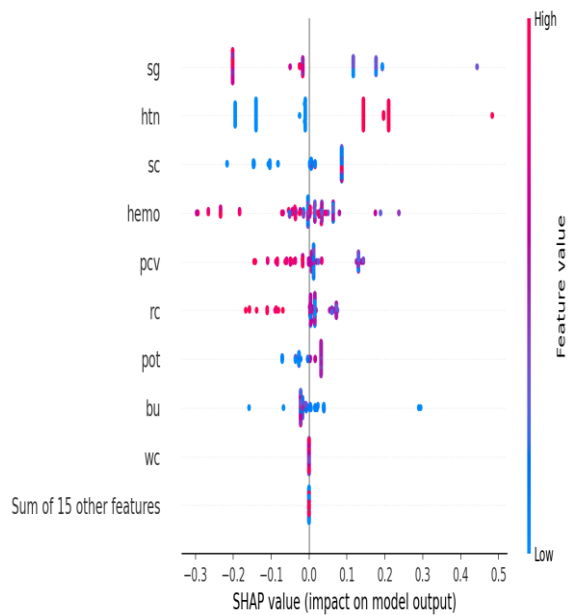


Fig 5: SHAP Summary Plot showing feature contribution to CKD prediction

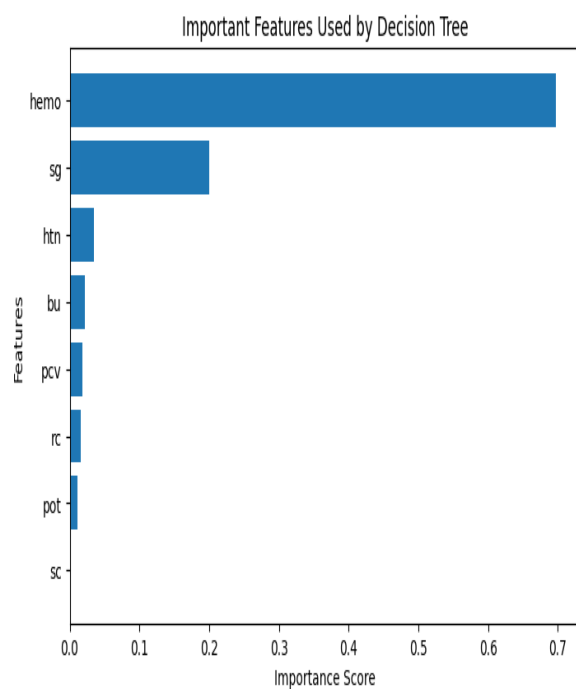


Fig 6: Global feature importance using SHAP values

The SHAP explainability framework improved transparency and interpretability by identifying how feature values positively or negatively influenced model predictions.

9. DISCUSSION

The experimental results demonstrate that explainability is essential in healthcare AI systems. Although predictive accuracy is important, healthcare professionals must also understand the reasoning behind predictions.

The integration of SHAP explainability techniques helped:

- Improve transparency in prediction outcomes
- Identify clinically important biomarkers
- Support trustworthy healthcare AI systems
- Improve interpretability in clinical environments

The findings suggest that explainable machine learning can assist healthcare professionals in making informed and transparent clinical decisions.

Despite promising predictive performance, the study has certain limitations. The healthcare datasets used may contain class imbalance and missing clinical attributes. In addition, machine learning models trained on limited datasets may face generalization challenges when applied to real-world hospital environments. Future studies should incorporate larger multi-center clinical datasets for improved robustness and external validation.

10. CONCLUSION

This research proposed an explainable machine learning framework for Chronic Kidney Disease prediction using a Decision Tree classifier integrated with SHAP-based interpretability analysis.

The model achieved strong predictive performance while maintaining transparency and interpretability through SHAP explainability techniques. Important healthcare features including Hemoglobin, Specific Gravity, Blood Urea, and Hypertension were

identified as significant contributors influencing CKD prediction outcomes.

The study demonstrates that Explainable AI can improve trustworthiness and transparency in healthcare prediction systems, supporting more reliable clinical decision-making processes.

11. FUTURE WORK

- Multi-model healthcare comparison
- Integration of SHAP and LIME explainability
- Deep learning-based healthcare prediction systems
- Multi-disease explainable AI frameworks
- Clinical deployment studies
- Real-time healthcare monitoring systems

12. REFERENCES

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